Answers to Computer Systems Workbook

1 Data representation

1 a 165 = 1 0 1 0 0 1 0 1

b 165 = A 5

2 a 0 1 1 1 0 1 1 0 = 118

b 5 F = 95

c 1 0 1 0 1 1 0 0 0 1 0 0 = A C 4

d 3 E D = (0 0) 1 1 1 1 1 0 1 1 0 1

3 a 800 × 16 = 12 800 MiB

b 12 800/1024 = 12.5 GiB

4 a i 65 = 0 1 0 0 0 0 0 1

ii 97 = 0 1 1 0 0 0 0 1

b i V = 86

ii v = 118

c i 86 = 0 1 0 1 0 1 1 0

ii 118 = 0 1 1 1 0 1 1 0

d Lower-case letters have an ASCII code value which is + 32 when compared to the equivalent upper-case letter, therefore to convert from an upper-case value, simply place a '1' in the column with value 32.

M = 77 = 0 1 0 0 1 1 0 1 and m = 77 + 32 = 109 = 0 1 1 …

5 a

0	0	1	0	0	0	0	1

b i

1	0	0	0	0	1	0	0

ii 132

c i R = 27 = 0 0 0 1 1 0 1 1

r = 108 = 0 1 1 0 1 1 0 0

ii M = 22 = 0 0 0 1 0 1 1 0

m = 88 = 0 1 0 1 1 0 0 0

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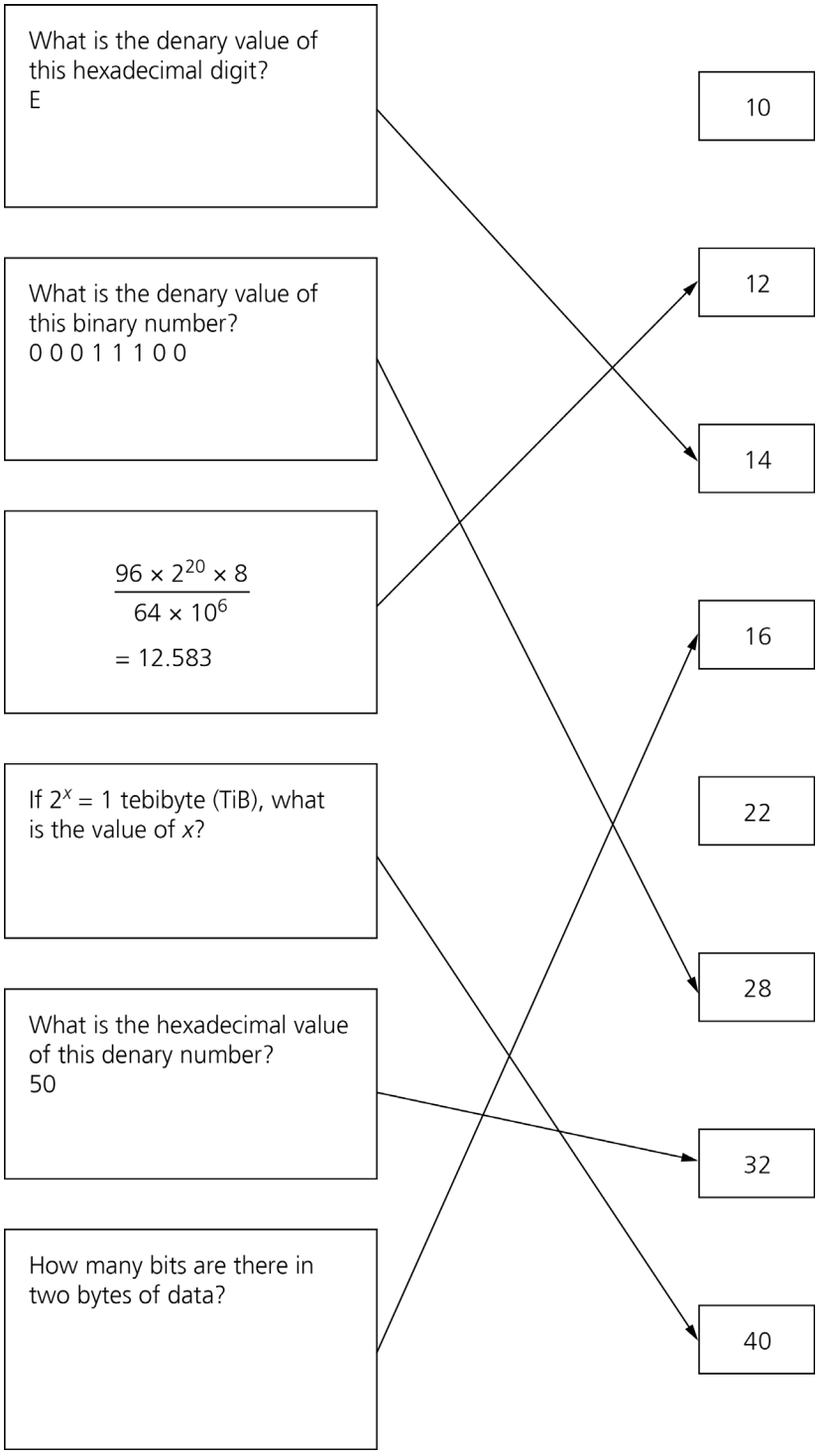
6 **a** 44801 = A F 0 1

b 1 0 1 0 1 1 1 1 0 0 0 0 0 0 0 1

7 Three from:

- memory dumps
- error codes
- MAC addresses
- IP addresses
- HTML colour codes

8



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9 a i

Number	X	D	OUTPUT
220	220	128	
	92		1
		64	
	28		1
		32	
	−4		0
	28		
		16	
	12		1
		8	
	4		1
		4	
	0		1
		2	
	−2		0
	0		
		1	
	−1		0
	0		
		0	end

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ii

Number	X	D	OUTPUT
73	73	128	
	−55		0
	73		
		64	
	9		1
		32	
	−23		0
	9		
		16	
	−7		0
	9		
		8	
	1		1
		4	
	−3		0
	1		
		2	
	−1		0
	1		
		1	
	0		1
		0	end

b The flowchart is converting denary (base 10) numbers into 8-bit binary (base 2) numbers.

10 a i 94

ii 61

b i 1 0 0 1 1 0 1 1

ii 155

This is equal to 94 + 61 in denary.

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11 a largest number = 1 1 1 1 1 1 1 1 = 255

b i 0 1 1 1 0 1 1 1 + 1 0 0 1 1 1 0 1 = 1 0 0 0 1 0 1 0 0

ii A ninth bit is generated therefore overflow error has occurred (i.e. the result exceeded the largest possible number which can be stored using 8 bits).

12 a 32 + 16 + 8 + 4 = 60

b i 0 0 0 1 1 1 1 0

ii denary value = 30

iii 0 0 0 0 0 1 1 1 – result is 7 which is incorrect (30 ÷ 4 = 7.5); 1-bit has been lost

c i 1 1 1 1 0 0 0 0

ii Value is 240 which is equivalent to 60×2^2 – this is because each logical shift left multiplies the original binary number by 2. Two shifts left multiplies by $2^2 = 4$.

13 a 37 = 0 0 1 0 0 1 0 1 and 19 = 0 0 0 1 0 0 1 1

b 0 0 1 1 1 0 0 0

c 1 1 1 0 0 0 0 0; denary value of (b) is 56; after logical shift, value is $56 \times 2^2 = 224$

d 0 0 0 0 0 0 1 1 – value after shift is 3 which is incorrect since $56 \div 2^4 = 3.5$; a 1-bit has been lost

14 a −128

b i Most negative is 1 0 0 0 0 0 0 0 = −128

ii Most positive is 0 1 1 1 1 1 1 1 = +127

c i 47 = 0 0 1 0 1 1 1 1

ii −59 = 1 1 0 0 0 1 0 1

d i 1 1 0 0 1 1 1 0 = −50

ii 1 1 1 1 1 1 1 0 = −2

e i 45 = 0 0 1 0 1 1 0 1

ii −45 = 1 1 0 1 0 0 1 1

iii 1 0 0 0 0 0 0 0 0 – in two's complement the final carry to the 9th bit does **not** signify overflow. We ignore this bit and consider the other 8 bits, which do give the correct answer of zero.

15 a (4096 × 2048 × 16)/(8 × 1024 × 1024) = 16 MiB

4096 × 2048

× 16 and ÷ 8

divide by 1024 × 1024

16 MiB

b size of sample = (number of samples per sec) × (number of bits per sample) × (length of sample size in secs) = (40 960 × 16 × 1024) bits

multiply by 2 for stereo recording and divide by 8 to get number of bytes

divide by (1024 × 1024)

to give 160 MiB

(40 960 × 16 × 2 × 1024)/(8 × 1024 × 1024) = 160 MiB

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16 a It saves storage space on the HDD/SSD.

It is faster to upload/download files to/from the internet.

It reduces the file size which allows the file to be emailed.

b Lossless – all data from the original file can be reconstructed when the file is uncompressed again.

Lossy – the compression algorithm eliminates unnecessary data from the original file; the original file cannot be reconstructed once it has been compressed.

c i mpeg-3, mpeg-4 or jpeg

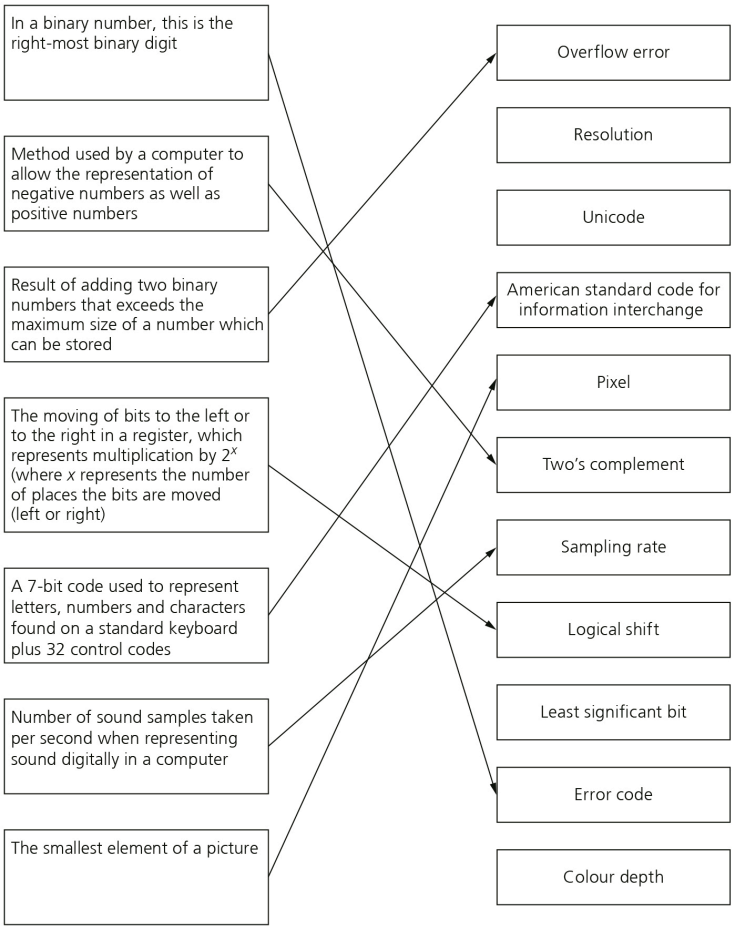
ii RLE (run-length encoding)

17 a colour depth – the number of bits used to represent each colour

b image resolution – the number of pixels that make up an image, for example, 4096 × 3192

18
- Standard ASCII code consists of 7-bit codes.
- It represents letters, numbers and characters found on a standard keyboard + 32 control characters.
- Extended ASCII code uses 8-bit codes, which allows for non-English characters.
- Unicode can be used to represent all languages of the world.
- Unicode overlaps with ASCII code since the first 128 characters are the same, but Unicode can support several thousand different characters.
- ASCII code uses one byte to represent a character, whereas Unicode will support up to four bytes per character.

19



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20 a B

b D

c A

d D

e B

21 a It is a form of lossless file compression:

- It reduces the size of a string of adjacent, identical data.
- A repeated string is encoded into two values:
  - The first value represents the number of identical data items.
  - The second value represents the code of the data item.
- RLE is only effective where there is a long run of repeated units.

b i 3W 2D 5W 4D 3W 1D 1W 2D 1W 1D 1W 16D 1W 1D 1W 2D 1W 1D 3W 4D 5W 2D 3W

ii 8 × 8 grid requires 64 bytes of storage

46 bytes are used in the RLE code

reduction = 64 − 46 = 18 bytes

22 a i the number of bits used to represent sound amplitude

ii the number of sound samples taken per second

b i 4 bits (15 = 1111)

ii 10 Hz (i.e. 10 samples per second)

iii point 9: (14) = 1 1 1 0

point 18: (11) = 1 0 1 1

c Benefits:

- larger dynamic range
- better/more faithful sound quality
- less sound distortion

Drawbacks:

- produces larger files
- takes longer to transmit/download music files
- requires greater processing power

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